INTELLIGENT AYURVEDIC HERBAL PLANT IDENTIFICATION AND ECONOMIC ANALYSIS IN SRI LANKA

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Project Proposal Report

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DECLARATION

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Ayurvedic medicine is a traditional method for treating diseases and disorders that humans have. Medicine herbals play a critical role in it. Ayurvedic medicine relies on quality assessment of medicinal plants, in which treatment efficacy relies on accurate identification of healthy and mature plant materials [1]. Nonetheless, the subjective quality evaluation that is done manually is subject to inconsistencies [1]. In the proposed research, an Explainable AI (XAI) system will be suggested to automate the quality assessment of Ayurvedic medicinal plants based on the image analysis method.

Although the deep learning models have demonstrated a high level of accuracy in classifying plants, their black-box nature is a critical research shortcoming, thus restricting their use in sensitive medical fields where transparency is critical [2]. This paper fills this gap by incorporating Grad-CAM (Gradient-weighted Class Activation Mapping) to demonstrate the decision of the model visually.

The solution offered is based on the use of the ResNet50 architecture through the transfer learning technique to classify plant leaves and fruits image into healthy, diseased, or overmature. The system is developed in Python, PyTorch, and OpenCV and produces heatmaps to indicate the location of specific parts of the plant, including necrotic lesions or maturity signs, that were used to make the classification.
The anticipated results are high-performance of deep learning model, a newly labeled dataset of Ayurvedic medicinal plant images, and a strong visual explanation facility. This system will help to improve transparency in the herbal supply chain, increase quality control, and build trust with Ayurvedic practitioners and researchers by increasing the transparency of AI-based evaluations [3].

Keywords – Ayurvedic Medicinal plants, Deep Learning, Quality Assessment, ResNet50, Grad-CAM

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LIST OF ABBREVIATIONS

Abbreviation	Full form
ML	Machine Learning
AI	Artificial Intelligence
DB	Database
F1 Score	Harmonic Mean of Precision and Recall

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1. INTRODUCTION

1.1 Background and Context

Traditional medicine (TM) is one of the primary healthcare curative options in the world, with its accessibility, affordability and strong cultural acceptance [1]. In places such as Africa, TM is used by up to 80% of the population, In regions like Asia and Latin America, the people are guided by historical and cultural beliefs to use it [1]. Ayurveda, a complete system of medicine that originated thousands of years ago in the Indian subcontinent, is one of the key elements of this tradition as it deals with the herbal and non-toxic plants to treat different diseases [4] [5]. Sri Lanka has its own Ayurvedic culture, which has been passed down through generations for more than 3,000 years and contains a large array of herbal medicinal plants [6] [4] [7].

The Ayurvedic treatments rely mostly on the correct identification and quality of medicinal plants [4] [5]. Nevertheless, confusion by morphological similarity between species and confusion with nomenclature are severe issues in the real world that lead to misidentification [1]. Studies have shown that out of the 30% of medicinal plants used in the classical literature, some plants have led to controversy over their real botanical names [1]. Manual identification is a cumbersome procedure that requires a lot of expert interference and it also possesses a probability of human error, especially when it concerns rare species or slight changes in quality [8].

To overcome them, deep-learning-based methods, namely Convolutional Neural Networks or CNNs, have become potent to use to automate the identification of plants and their quality [3] [4]. Although these models have been able to attain impressive classification rates (up to 95%), they are more of a “black-box” [2] [9] [10]. Such an inability to interpret is a major research gap; agricultural specialists and Ayurvedic practitioners cannot trust or rely on AI decisions when the reasoning is not transparent [2] [10]. The focus on accuracy is also common in current studies, and the validation of acquired characteristics is not considered, which prevents the application of AI in sensitive fields of medicine [2].

The above limitations are tackled in this proposed research, which will build an Explainable AI (XAI) system to evaluate the quality of Ayurvedic plants. The system will classify the condition of plants (healthy, diseased, or overmature) by using a ResNet50 architecture, but will also produce heatmaps to explain to the user visually what part of the plant the model has relied on to make a specific decision. The project is consistent with SDG 3 (Good Health and Well-being) as they are guaranteed with the integrity of the herbal medicines and belongs to the Centre of Excellence for AI (CoEAI) research cluster, with a Technology vertical and a Healthcare one.

Literature Review

Recent academic studies have paid a growing interested to the automated identification and quality analysis of medicinal plants through computer vision and especially as the application of traditional medicine (TM) is rapidly expanding around the world [1] [11]. This literature review analyzes the current deep learning strategies, mobile-based products, and the new combination of Explainability in Artificial Intelligence (XAI).

Identification and Classification Systems A large part of present studies is concerned with species recognition. Lankasena et al. (2024) solved the confusion of misidentification of Ayurvedic plants, including those belonging to the genus Bauhinia, with a CNN EfficientNet-B0 model, with a validation accuracy of 96.16% [1]. On the same note, Jayanka and Fernando (2020) used VGG-16 to identify 17 types of the Ayurvedic herbal plants in Sri Lanka, stating that deep learning models yield much more significant results when compared to traditional Multi-layer Perceptrons (MLP) [5] [4]. Pathiraneg et al. created the Arogya platform that uses VGG-16 with an accuracy of 99.53% to classify and a Marker-based Watershed algorithm to detect plants [4]. Moreover, Senevirathne et al. (2020) introduced an assistant that does not only recognize the plants with a 92.5% hit rate but also supplies the model in 3D in the Augmented Reality (AR) to improve the learning process of the user [7].

Quality and Disease Detection, other than identification, researchers have investigated quality grading. In the article by Zulkifli et al. (2025), a quality detection system based on peel analysis was created, where a CNN was used to differentiate between “good” and “bad” samples of lemon [3]. Haque et al. (2025) proposed a lightweight CNN framework in the field of classification of potato leaf disease, with 99.14% accuracy [2]. They stressed that the discovery of such diseases as Early and Late Blight is a critical factor in agricultural productivity [2]. To reduce the knowledge gap among the Ayurvedic practitioners, Gamage et al. (2024) suggested an integrated system employing VGG16 (96.88% accuracy) with multilingual functionality and social media crowdsourcing [6].

Explainable AI (XAI) and Grad-CAM Applications, the progression of making black-box models to transparent models is an emerging research field. Yadav et al. introduced a hybrid model designed specifically to detect medicinal plants with the help of Grad-CAM visualization to identify biological attributes [12]. Gopalan et al. (2025) applied a ResNet152 with Grad-CAM to detect corn leaf disease due to the argument that farmers cannot be confident in using models such as YOLOv8 or DenseViT since they do not answer the question of “how” a prediction was reached [10]. They showed that heatmaps proved to be effective in showing the areas of the leaf that are the most important to the model, e.g., a necrotic lesion or common rust [10].

Limitations of Existing Systems, regardless of high accuracy, there are a number of gaps:

- **Interpretability:** The vast majority of state-of-the-art models (e.g., DenseViT, YOLOv5) are more about accuracy and not transparency, so users cannot validate learned features [10] [2].
- **Robustness:** Lightweight models such as MobileNetV3 tend to be trained on synthetic data sets and fail to run in the real-world farm setting where the lighting and backgrounds differ significantly [10] [2].
- **Domain Specificity:** There are general plant identification applications such as "Pl@ntNet" and "PlantSnap", but they do not always recognize endemic plants of Sri Lanka or even particular measures of Ayurvedic quality [7] [6].
- **Consistency:** spread explanations, such as Layer-wise Relevance Propagation (LRP), can be used to carry out segmentation, but may be inconsistent between diseases [10].

1.3 Research Gap

Although there exists remarkable progress in the application of deep learning to analyze agricultural and medicinal plants, there are still some critical research gaps, which impede the successful use of the technologies in the Ayurvedic industry:

- **Lack of Interpretability and Trust:** One of the biggest drawbacks of the state-of-the-art models (i.e. DenseViT, YOLOv8, and custom CNNs) is that they are "black-box" [2] [10]. These systems normally generate a classification label which could be diseased or healthy without providing the reasoning behind the prediction [2]. To Ayurvedic practitioners and researchers, this lack of transparency is a crucial obstacle; they are not able to have complete confidence in AI-based evaluations of the quality of the medicines when they cannot discover what biological aspects the model is actually prioritizing [2] [10].
- **Insufficient Robustness in Real-World Conditions:** Most of the current plant classification models are trained on clean, laboratory-captured datasets such as PlantVillage [2]. Although these models are very accurate in controlled settings, they usually do not generalize to real-life situations where the lighting, camera angles and the background noise are much different [2] [10]. Moreover, models such as MobileNetV3 can be trained on synthetic data, and this can be seen as a limitation to their use in a natural farm environment [2] [10].
- **Narrow Focus and Dataset Limitations:** The bulk of the existing literature is concerned with a broad range of plants or typical farm crops (such as corn or potato), which does not take into account the morphological complexities of

Ayurvedic medicines [1] [2]. Research has seldom provided a detailed examination into the ability to differentiate morphologically close medicinal leaves which is the root of most confusion in herbal medicine [1]. About 30% of the medicinal plants described in classical literature are still disputed with respect to their botanical names, but AI applications specifically to local occurrences have been relatively ignored [1].

- **Static Nature of Models:** Existing systems can be very static; once they have been trained, they fail to dynamically adjust to new or changing data without human intervention [9]. This renders them ineffective in detection of any new plant disease outbreak or in the modification of quality standards when new information is obtained [9].
- **Absence of Domain-Specific Integrated Solutions:** Although there are general identification applications, they often cannot recognize endemic medicinal species in Sri Lanka or even specific Ayurvedic quality indicators [6]. A significant lack of centralized, smart systems adapted to serve the Ayurvedic industry with high-precision quality analysis and explainable feedback and local language support can be identified [6] [4].

This study addresses these gaps by proposing an Explainable AI (XAI) system using Grad-CAM to give visual explanations to its quality measures [2]. By emphasizing biologically significant features - like necrotic lesions or maturity indicators - the system helps instill trust and makes sure that the model is gaining knowledge on actual plant patterns instead of being biased by some background information [2] [10] [12]. This transparency and accuracy integration is a novel approach to ensuring the integrity of the herbal supply chain in Sri Lanka.

2. OBJECTIVE

2.1 Main Objective

The primary goal of this study is to design an Explainable Artificial Intelligence (XAI) system that automatically analyzes the quality of Ayurvedic medicinal plants through image analysis based on deep learning, namely, a ResNet50 architecture and Grad-CAM visualisation to make proper predictions regarding their suitability and provide visual explanations for its predictions.

2.2 Specific Objectives

The particular goals of this research part are as follows:

To produce a powerful deep learning model that can accurately categorize Ayurvedic medicinal plant materials, such as leaf and fruit, into quality classes, including, but not limited to, healthy, diseased, or overmature.

To use a transfer learning method with a pre-trained ResNet50 architecture in order to guarantee a high level of classification and effective feature extraction even in case of a narrow or specialized medicinal plant dataset.

To integrate Grad-CAM (Gradient-weighted Class Activation Mapping) visualization to offer post-hoc visual explanations on model predictions, highlighting the particular biological regions- e.g. necrotic lesions or signs of maturity- that have been used to make the decision.

To perform an empirical evaluation of the system's performance in terms of standard measures of accuracy, precision, recall and F1-score to confirm the effectiveness and reliability of the system in terms of clinical or supply-chain application.

To provide a transparent, intelligent assistant system that can help Ayurvedic practitioners, herbal vendors and non-experts to find and assess the quality of plant substances with great confidence and comprehensible feedback.

3. METHODOLOGY

This element of the research is based on an Experimental / Empirical Research Methodology. The reason behind the choice of this approach is that the main objective is to compare the performance of deep learning models and confirm visual explanation performance in controlled experimental conditions. The methodology allows the rigorous evaluation of classification accuracy and model interpretability because of the use of both quantitative metrics and qualitative visual assessment.

3.1 Algorithms and Technical Methods.

The system suggested works with a deep learning pipeline, which consists of multi-stages:

- **Transfer Learning based Feature Extraction:** The system will use a ResNet50 architecture that has been trained on ImageNet. Transfer learning is applied to exploit prior information of visual patterns (edges, textures, shapes), which is especially successful when special medicinal plant datasets are applied.
- **Classification:** The last fully connected layers of the ResNet50 model will be changed to classify the images of medicinal plants into three different classes of quality: Healthy/Suitable, Diseased, and Overmature.
- **Grad-CAM based explainable AI (XAI):** To facilitate the transparency, Gradient-weighted Class Activation Mapping (Grad-CAM) will also be incorporated. The method determines gradients to generate heatmaps, which indicate the particular biological areas, including necrotic lesions or maturity indicators, which have the greatest effect on the model prediction [12].

3.2 Tools and Platforms

The technologies that were chosen according to their strength and usage in the industry are the following:

Python: Python is selected as the main programming language because it has a large ecosystem of AI and data science.

- **PyTorch:** It is the fundamental deep learning platform used in the creation and optimization of models.
- **OpenCV:** This is used in essential computer vision activities, such as image normalization and noise elimination [10].
- **Google Colab / GPU Platforms:** This will be utilized to offer the required calculation power to run deep convolutional neural networks.

3.3 Data Collection and Preprocessing

- **Data Acquisition:** Images of Ayurvedic medicinal plant leaves and fruits will be collected via field photography, visits to herbal farms, and the use of publicly available datasets
- **Labeling:** The domain experts will manually label each image in the predefined quality categories.
- **Preprocessing:** Image will be rescaled to a fixed resolution (e.g. 224x224) as well as normalized to achieve consistency in model convergence [10] [12].
- **Augmentation:** Data augmentation methods will be used to enhance the generalization process and avoid overfitting: flipping, rotation, and brightness adjustments [10].

- **Data Splitting:** The data will be separated into training and testing data by splitting it into 80:20 portions to provide a balanced assessment [12] [13].

3.4 Strategy and Performance Metrics of Validation.

The validity of the system will be tested on a two-pronged basis:

- **Quantitative Evaluation:** It will be evaluated based on standard measures, such as Accuracy, Precision, Recall, and F1-score [10] [13].
- **Qualitative XAI Analysis:** The Grad-CAM heatmaps will be inspected visually to make sure that they are connected to relevant biological indicators of plant quality [10]. This move confirms that the model is basing its decisions on pertinent features as opposed to background noise [2] [12].

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overall Solution Overview

The proposed system will be an architecture based on three tiers (Frontend User Interface), Backend Processing Unit, and an AI Engine with an Explainability (XAI) module [3]. The system enables automated evaluation of the quality of Ayurvedic medicinal plants automatically with image data turned into a classification of healthy, diseased, or overmature plants with visual heat maps to support the choice of the model. This architecture will guarantee a smooth transition between the data collection and delivery of clear results [3] [4] [14].

4.2 System Diagram

Its architecture is based on a sequence of a pipeline whereby user-supplied images are run through deep learning layers that identify features and produce predictions [3] [2] [10].

- **User Layer:** This is available through an interface mobile whereby the user will take pictures or upload pictures of plant leaves or fruits [4] [8] [9].
- **Processing Layer:** A back-end framework (e.g., Flask) to handle API requests, image normalization, and communication between the back end and the AI model [3] [9] [14].
- **AI Engine (ResNet50):** This is the fundamental unit that employs the deep-residual learning to detect intricate plant patterns and disease indicators [12] [9] [10].
- **XAI Module (Grad-CAM):** This is a post-hoc analysis layer that calculates gradients to generate heatmaps that are then superimposed on the original image to visualize the parts of the image that have an effect [12] [10].

4.3 Interactions between the Components and Data Flow.

The system has a set of stages of data flow, which are interdependent:

- **Image Acquisition:** The end user enters an RGB image of a part of a medicinal plant into the front end [14] [3].
- **Preprocessing:** The backend processes the image to a standardised 224x224 size and provides normalisation so that the ResNet50 model can be consistent [9] [10].
- **Feature Extraction & Classification:** The processed image is inputted into the ResNet50 architecture, and structural features are detected by residual block using hierarchical convolutional layers [10] [12]. An ultimate Softmax layer is used to give the probabilities to the quality types [2] [12] [14].
- **Grad-CAM Heatmap Generation:** At the same time, the system calculates the gradients of the predicted class score with respect to the final convolutional layer [12] [10]. These gradients can be used to produce a heatmap to indicate biologically significant features such as necrotic lesions or maturity spots [2] [12].
- **Output Delivery:** The end forecast is returned to the frontend and the heatmap of the explanations is displayed to the user [3] [14].

4.4 Scalability and security issues.

In order to achieve high performance and high reliability of the system, questionnaires will be integrated into the system according to the following considerations:

- **Scalability:** The backend can be containerized with the help of Docker and can be deployed to such cloud providers as AWS, thus the system may address the growing number of user requests and the number of incoming requests at the same time [3] [14].
- **Security:** RESTful APIs are secure and provide an authorized communication between the client and the server [9]. RBAC may be deployed to allow only expert administrators to do sensitive tasks, including dataset management or retraining models [9].
- **Data Integrity:** A structured database Firebase will be used to ensure that user interactions and history of classification is saved in an efficient way and will be reviewed in the future to audit the model and improve it [8] [9].

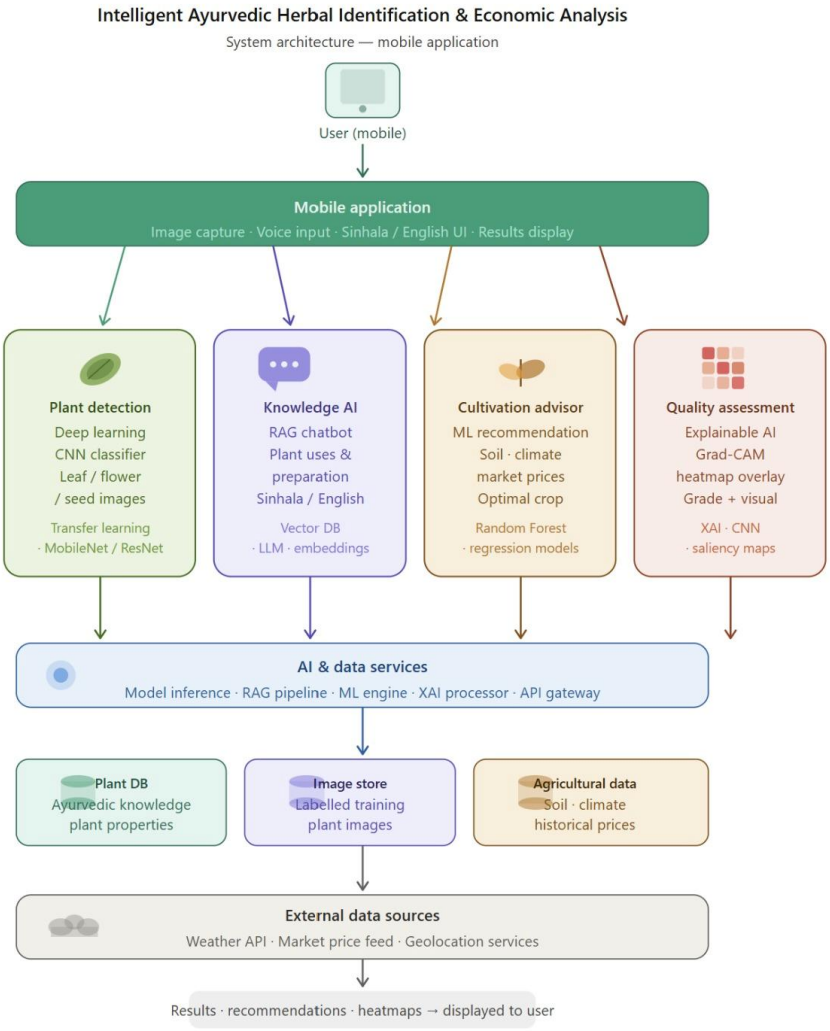


Figure 1:system architecture

5. User Requirements

5.1 Stakeholder Identification The key stakeholders of the Explainable AI (XAI) system are:

- **Ayurvedic Practitioners and Researchers:** The professionals that need high-confidence quality measurements in medicinal formulations.
- **Farmers and Herbal Suppliers:** The users in the rural or agricultural setting that require to detect the quality of plants promptly to minimize wastage and maintain integrity in the supply chains [14].
- **Novice Users:** Users, who are yet to be familiar with the medicinal herbs and need to be provided with an intuitive and interpretable feedback to help them identify them [9].

- **Domain Experts (Admins):** Plant pathologists or senior Ayurvedic experts who will label the images that are not known to fines the accuracy of the model in the long run [9].

5.2 Functional Requirements The system will support the following functionalities to support the needs of the users:

1 **Image Acquisition:** The system will enable users to take or post RGB images of plant leaves or fruits of medicinal plants through mobile [14].
2 **Automated Quality Classification:** The system will classify uploaded images based on a ResNet50 model, which will be used to classify plant materials in three states, namely, Healthy/Suitable, Diseased, or Overmature.
3 **Explainable Visual Justification:** In each classification, the system will produce Grad-CAM heatmaps to show the particular biological components (e.g., maturity spots or lesions) that were used to guide the decision of the model.
4 **Result Visualization:** The system will be able to show the final prediction, a confidence score, and the explanatory heatmap through the user-friendly graphical interface [9] [14].
5 **Subscription-Based User-Access:** The system has a simplified access control model that only has standard users. Only the users who have made a successful payment or have an active subscription will be able to access the core functionalities, i.e., uploading images, running the deep learning analysis, and seeing the automated quality predictions and Grad-CAM visual explanations.

5.3 Non-Functional Requirements To guarantee reliability and trust, the system should follow up to the following quality attributes:

- **Accuracy:** The deep learning model should have high metrics of classification performance such as precision and F1-score to be useful in medicine [8].
- **Interpretability:** The visual descriptions (heatmaps) should be sufficiently understandable so that non-experts can identify biologically significant characteristics and random noise [3].
- **Usability:** It should be user-friendly and need less technical skill to make it applicable to farmers and practitioners in the field [9] [14].
- **Performance:** The system will support real-time field analysis by having quick response times to make predictions as well as heatmap [9] [8] [14].
- **Scalability:** The architecture should be in a position to process large-scale image data and be able to support the introduction of other Ayurvedic plant species into the knowledge base [9].
- **Security:** the frontend and the backend should be secured through APIs, and user data as well as model parameters should not be accessed without the proper authorization [9].

- **Reliability:** The system should be consistent in different environmental situations like in cases of different lighting conditions or background complexities [14].

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona The target market of this system will include traditional medicine industry and more specifically it will focus on Ayurvedic practitioners, herbal medicine factories, research and local farmers who grow and sell medicinal plants..

- **Customer Persona:** A quality assurance manager of an Ayurvedic pharmaceutics or a practicing Ayurvedic doctor. They must quickly and precisely examine the health and maturity of medicinal leaves or fruits on the one hand to guarantee the effectiveness of their treatments but they frequently have no time or extremely specialized botanical expertise to do so by eye..

Value Proposition At present, the quality check on Ayurvedic plants is a laborious process that is time consuming and extremely prone to human error and subjectivity.. The suggested system is an automated and stable alternative to classifying plant materials (e.g., healthy, diseased, overmature) with a high degree of accuracy on the basis of deep learning.. More importantly, it addresses the issue of the black-box of standard AI by using Grad-CAM to produce visual heatmaps.. This Explainable AI (XAI) option shows in detail what elements in the leaf or fruit the model relied on to make a decision, which will create understandable evidence that makes the medical professionals and farmers trust it..

Revenue Model In line with the role-based access design of the system, the basic revenue stream will be a Subscription-Based Model (Software-as-a-Service). The core functionalities will be necessary to allow users to run the deep learning analysis and see the Grad-CAM visual explanations, which will be paid as a recurring subscription fee (monthly or annually). Moreover, a B2B Licensing Model might be implemented regarding the large Ayurvedic manufacturing firms, and they could incorporate the API of the system in their pipeline of quality-control in the industrial facilities.

Cost Estimation The start-up cost of the development should include the cost of high-performance computing (high-performance computing or Google Colab Pro or AWS instances) to train the ResNet50 architecture.The cost of traveling to herbal farms to collect data, purchase image annotation software, and the purchase of a secure data storage.. The commercialization phase will entail operational expenses such as continued hosting of the cloud server with the backend and database administration to process user subscriptions as well as the marketing cost to access the target population.

Pricing Stategy To attract the various market segments, tiered pricing approach will be employed:

Basic: Tier: The tier is aimed at individual farmers, beginner learners, or small-scale practitioners, by providing a fixed amount of image scans each month at a very affordable price.

Premium/Enterprise Tier: It is aimed at Ayurvedic hospitals, research institutions, and commercial herbal suppliers. This level will provide unlimited scanning daily, picture processing in large volume, been seen as a priority customer or client and a sophisticated dashboard to analyse quality trends over time.

Competitive Advantage Although there exist commercial generalized plant identification and agricultural disease detection software in the market, they are largely black boxes which require customers to blindly follow their forecast.. The main competitive strength of this system is that it is domain specific to Ayurvedic medicinal plants, as well as its Explainable AI (Grad-CAM) integration.. The system offers a certain degree of transparency and diagnosis confidence, through the visual justification of its evaluations, including pointing out particular necrotic lesions or signs of maturity, that can not be achieved by the currently competing systems..

Intellectual Property Considerations The quality Ayurvedic medicinal plant images newly labelled with high quality images with the annotations that are domain-specifically labelled by domain experts will be a highly valuable digital asset and be safeguarded as a trade secret.. The copyright law will secure the custom-trained ResNet50 model weights, the actual backend processing algorithms, as well as the application source code. Besides, the name, branding, and logo of the application will be registered as a trademark to create a competitive identity in the digital healthcare and AgriTech markets.

7. BUDGET AND JUSTIFICATION

Table 1 Budget Cost Estimation

Resource	Estimated Cost (Rs.)
High-performance GPUs (for training the ResNet50 architecture)	Rs. 0 (University / free Colab)
Data Collection and Site Visits (Travel to herbal farms and research institutes)	Rs. 5,000
Procurement of Image Annotation Tools and Data Storage	Rs. 11,000
Cloud Computing Services (e.g., AWS or Google Colab Pro for extended training)	Rs. 15,000
Total Estimated Budget	Rs. 31,000

Justification of Expenses

High-performance GPUs: The training of deep learning models, namely the ResNet50 architecture which has millions of parameters, is very computationally intense. This will be enabled majorly by the university resources or even by free versions of cloud services such as Google Colab.

Data Collection and Fieldwork: This requires traveling to collect data in an authentic and quality dataset by visiting institutions like Ayurveda Research Institute at Koswinna, Navinna or Peradeniya Herbal Garden. The use of direct field photography is necessary to ensure the capture of the variations of plants (healthy, diseased, overmature) under the conditions of the real-world light.

Annotation Tools and Data Storage: The process of annotating the condition of medicinal plants needs to be done manually by professionals. This labelling and high-capacity storage to handle large image datasets are some of the areas that the budget will be allocated.

Cloud Computing Services: Although the initial experiments are performed with free platforms, the training of more complex models and high-resolution Grad-CAM heatmaps could be considered premium cloud computing that can exclude waiting time and constraints imposed by hardware.

Licenses: Project will use Python, PyTorch and OpenCV which are open-source and hence do not require any licensing.

8. WORK BREAKDOWN STRUCTURE (WBS)

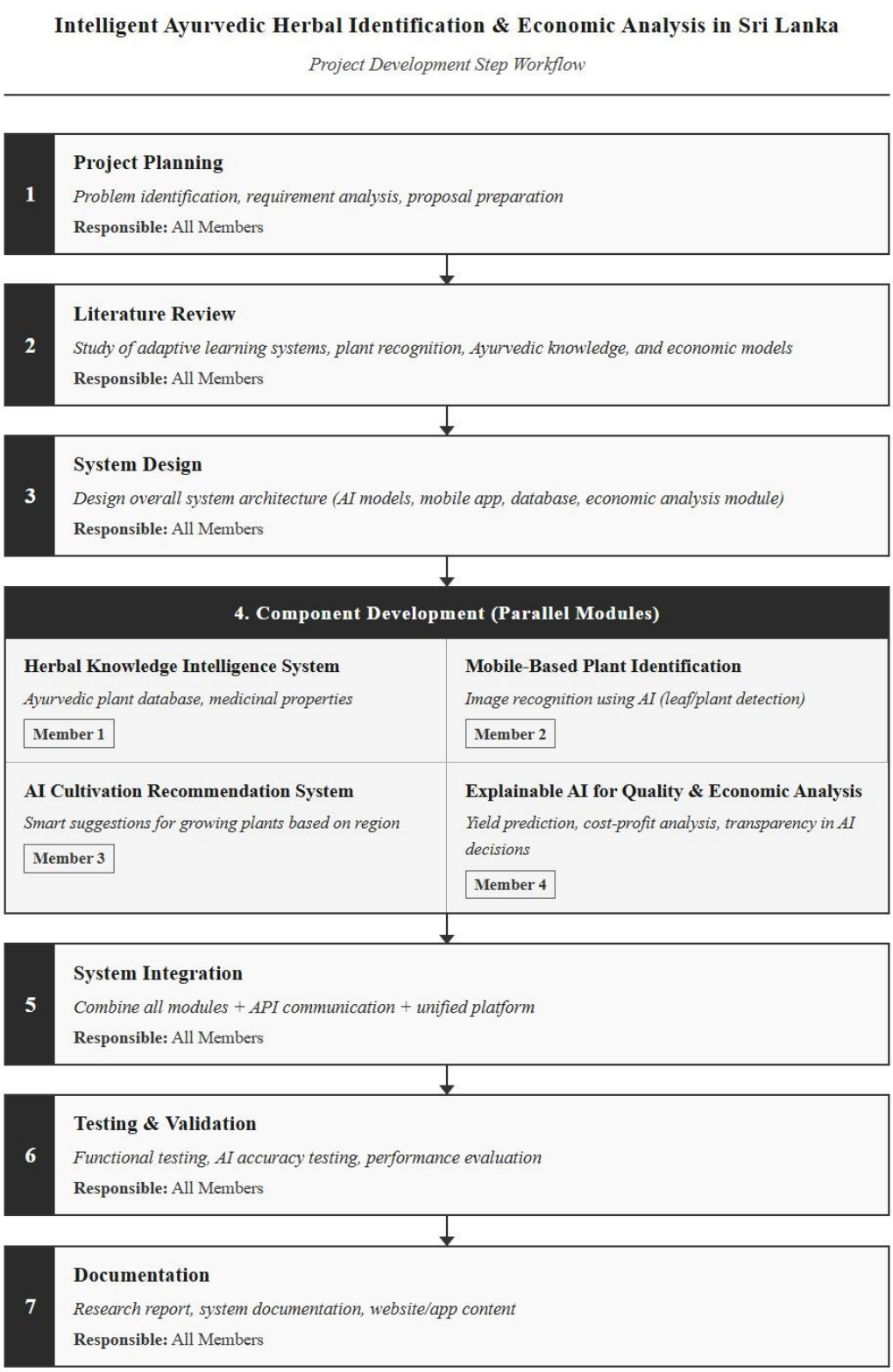


Figure 2:WORK BREAKDOWN STRUCTURE

9. GANTT CHART

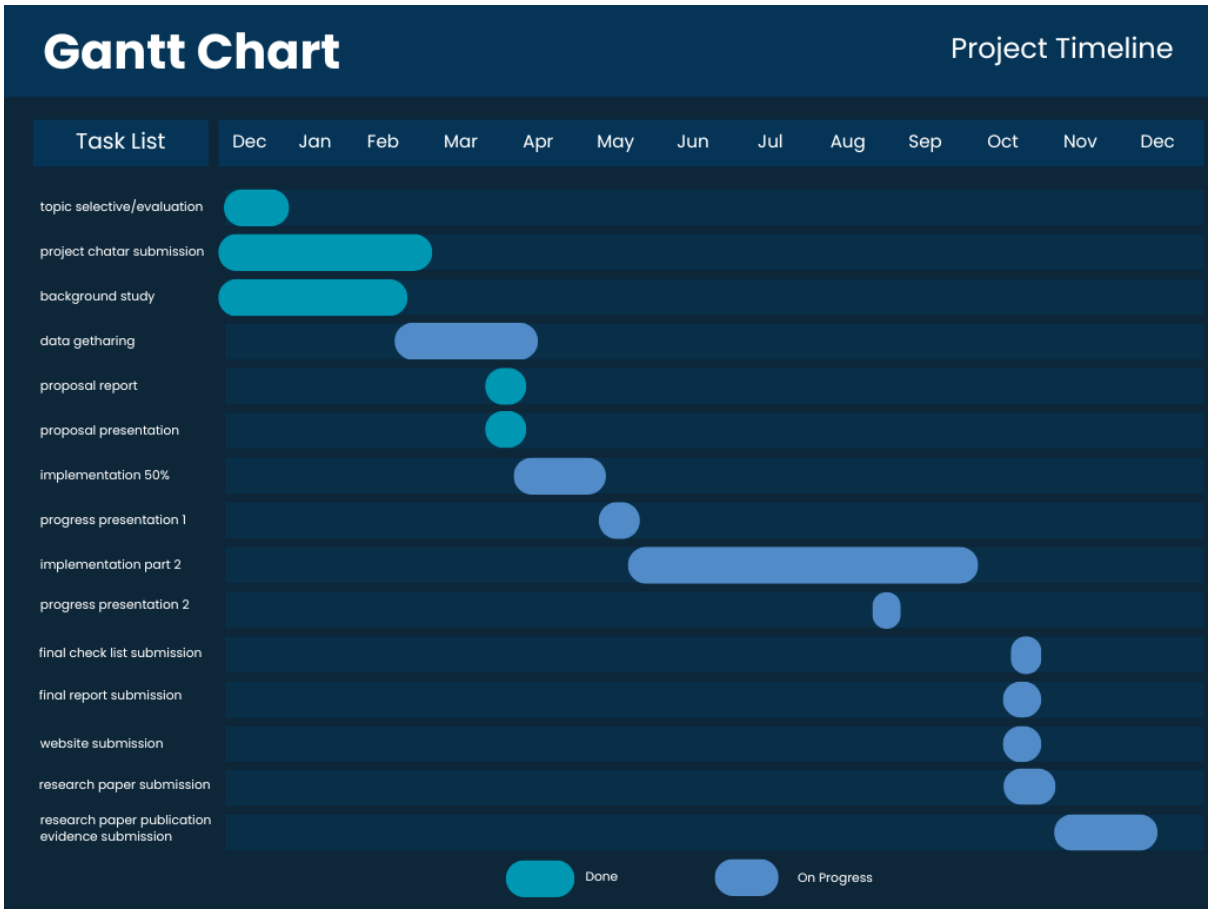


Figure 3:Gantt Chart

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[13] A lightweight and explainable CNN model for empowering plant disease diagnosis

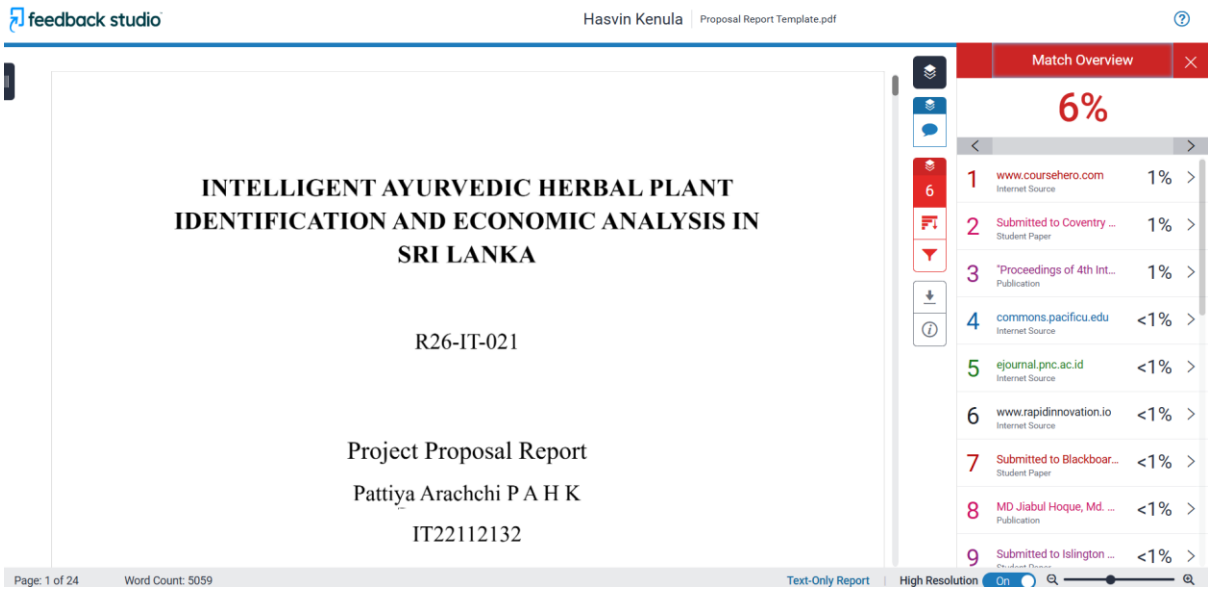
[14] AgriScan: Next.js powered cross-platform solution for automated plant disease diagnosis and crop health management

[15] Identification of Medicinal Plants by Visual Characteristics of Leaves and Flowers

[16] Construction of deep learning-based disease detection model in plants

[17] Machine Learning Methods in the Environmental Sciences: Neural Networks and

11.APPENDICES



APPENDICES 1:Turnitin screen shot